

Spotter Freight Rate Prediction

Machine Learning Engineer Assessment Report

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1. Executive Summary

This solution predicts freight load rates using historical shipment data with a time-aware validation strategy. The labeled development set contains 48,000 loads from January through October 2025, while the final validation set contains 12,000 later loads. Because the prediction period is in the future relative to the labeled data, chronological holdouts were used instead of random cross-validation.

The selected primary model is Ridge Regression with $\alpha=100$. It achieved the lowest mean MAE across rolling monthly backtests and remained more stable than nonlinear and feature-expanded alternatives. The final model was then trained on all 48,000 labeled rows and used to generate predictions for all 12,000 validation loads.

2. Data Exploration and Quality

Issue	Observed	Treatment
Missing weight	300 train; 165 validation	Training-median imputation + missing indicator
Missing market_index	374 train; 249 validation	Training-median imputation + missing indicator
Negative weight	292 train; 145 validation	Converted to absolute value as sign errors
Unseen locations	8 validation pickup/delivery cities unseen in train	One-hot encoding with unknown handling
Target distribution	Right-skewed; max observed rate \$25,533	Retained; no unsupported outlier deletion

Distance was the strongest numeric predictor in exploratory analysis (correlation with posted_rate approximately 0.91). The target was strongly right-skewed, so both MAE and RMSE were tracked to distinguish typical errors from rare large misses.

3. Validation Strategy

The development data spans 2025-01-01 through 2025-10-31. A random split would allow earlier and later market conditions to mix and would not match the real prediction setting. The first model-selection split therefore trained on January-September and held out October. Hyperparameter selection was then checked with rolling monthly holdouts for July, August, September, and October.

4. Model Experiments

Model	MAE (\$)	RMSE (\$)	R2
Ridge, alpha=100	140.46	651.30	0.8185
Enhanced Ridge	153.90	669.69	0.8081
HistGradientBoosting	162.62	663.18	0.8118
Naive median	1,146.79	1,567.97	-0.0522

HistGradientBoosting and a Ridge variant with lane and interaction features both underperformed the simpler Ridge baseline. This indicated that the dominant signal was largely linear and that added complexity was not improving future-month generalization.

5. Rolling Backtest and Final Selection

Validation month	Ridge alpha=100 MAE (\$)	Interpretation
2025-07	170.85	Higher-error month
2025-08	149.71	Improved
2025-09	140.94	Strong
2025-10	140.46	Strong

Across these four chronological folds, alpha=100 achieved mean MAE \$150.49, median MAE \$145.33, and mean RMSE \$632.92. It had the lowest mean MAE among the tested regularization strengths (1, 25, 50, 100, 200, and 500), so it was selected for the final model.

6. Error Analysis

On the October holdout, the median absolute error was \$61.33, the 90th percentile was \$193.37, and the 95th percentile was \$274.52. Only 62 of 4,853 rows had absolute error above \$1,000. This explains why MAE is low while RMSE remains much larger: a small number of extreme-rate loads dominate squared error.

7. Final Main Model

After model selection, the Ridge pipeline was retrained on all 48,000 labeled rows. Numeric features were standardized and categorical features were one-hot encoded with `handle_unknown='ignore'`. The final feature set included pickup/delivery coordinates, distance, weight, market_index, quote_signal, calendar features, a continuous days-since-start feature, and missing-value indicators, plus pickup, delivery, and equipment categories.

The final model generated exactly 12,000 validation predictions. All predictions were finite and positive, and the output IDs matched the provided validation template exactly. The prediction range was approximately \$133.69 to \$6,590.99, with mean \$2,424.03.

8. December Scenario

The fixed December input contains 31 rows for the same shipment: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb, with only the date changing. The file does not provide market_index, quote_signal, or coordinates, so these values were not fabricated. Instead, a separate reduced-feature Ridge model was trained using only information available in the December file.

The exact Lexington to Fort Wayne lane appears 32 times in the historical training data. A lane-aware reduced Ridge model slightly outperformed the non-lane reduced model in chronological backtesting (mean MAE \$204.46 vs. \$204.64), so the lane-aware version was selected. December predictions range from \$855.38 to \$914.92, with mean \$881.95.

9. December Prediction Chart

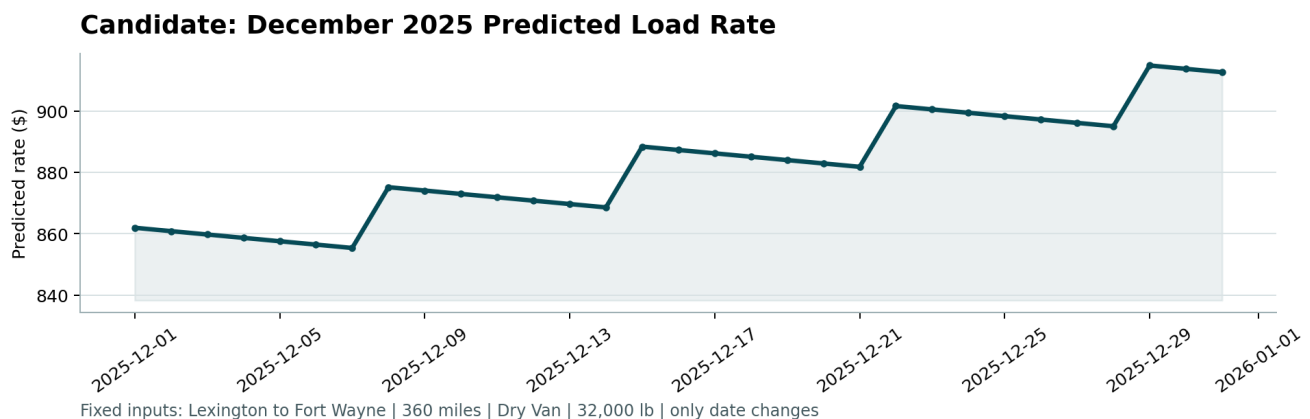


Figure 1. Fixed December 2025 prediction chart for Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb.

10. Reproducibility and Verification

The notebook workflow was converted into a reusable src/ pipeline. Running `python -m src.train` reproduces both the 12,000-row validation output and the 31-row December output. The reproduced predictions matched the originally validated files exactly, with maximum absolute difference 0.0 for both outputs.

Automated tests cover preprocessing, missing-value handling, feature definitions, leakage prevention, lane construction, prediction schema, duplicate-date checks, and rejection of non-positive predictions. All 13 tests passed.

The provided Spotter scorer was run against both the final submission files and the reproduced outputs. In both cases it validated all 12,000 final predictions, validated all 31 December predictions, and generated `candidate_december.png`. The scorer states that final hidden validation metrics are calculated by Spotter after submission.

11. Conclusion

A simple, regularized linear model provided the strongest and most stable future-month performance for this dataset. The final solution prioritizes temporal validation, training-only preprocessing statistics, safe handling of unseen categories, reproducibility, and strict submission validation. More complex models and interaction-heavy features were retained as documented experiments but were not selected because they did not improve chronological holdout performance.